综合练习

人员

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上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/2900

比赛概况 题目列表 选择题列表 提交记录 实时榜单 选择题排行榜 编辑比赛

王向东老师周日八点半C++对顶堆

刷新

#	用户名	姓名	编程分	时间	A	B	C
1	liuyichen	刘奕辰	300	3441	100	100	100
2	yangjunyan	杨俊彦	300	6824	100	100	100
3	lijinshu	李锦澍	300	6849	100	100	100
4	chenxinmiao	陈欣妙	300	10457	100	100	100
5	yuanchenjun	袁晨峻	200	1278	100	100	0

本周作业

https://cppoj.kids123code.com/contest/3156 (课上讲了 A ~ D 题, 课后作业是 D 题)

课堂表现

今天同学们课上做题表现很好, 提出表扬!

课堂内容

Max Multiple

dp, f[i][j][k]: 考虑前 i 个数, 选了 j 个, 和模 D 为 k 时, 最大和是多少

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 100 + 5;
int w[maxn], f[maxn][maxn][maxn];
// f[i][j][k]: 考虑前 i 个数, 选了 j 个, 和模 D 为 k 时, 最大和是多少

signed main()
{
  int n, K, D; cin >> n >> K >> D;
  for (int i = 1; i <= n; ++i) cin >> w[i];
```

```
  memset(f, -0x3f, sizeof(f)); f[0][0][0] = 0;
  for (int i = 1; i <= n; ++i) {
    for (int j = 0; j <= K; ++j) {
      for (int k = 0; k <= D-1; ++k) {
        f[i][j][k] = f[i-1][j][k];
        if (j) f[i][j][k] = max(f[i][j][k], f[i-1][j-1][(k-w[i]%D+D)%D] + w[i]);
      }
    }
  }

  if (f[n][K][0] >= 0) cout << f[n][K][0] << endl;
  else cout << -1 << endl;
  return 0;
}
```

Crystal Switches

分层图 + 01bfs

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 4e5 + 5;
const int inf = 0x3f3f3f3f;
struct node {
  int to, value;
};
vector<node> vec[maxn];
int dis[maxn];
bool st[maxn];
int n, m, K;

int bfs() {
  memset(dis, 0x3f, sizeof(dis));
  deque<int> dq; dq.push_back(1); dis[1] = 0;
  while (!dq.empty()) {
    int u = dq.front(); dq.pop_front();
    if (st[u]) continue;
    st[u] = true;

    for (node it : vec[u]) {
      int to = it.to, value = it.value;
      if (dis[u]+value < dis[to]) {
        dis[to] = dis[u] + value;
        if (value == 1) dq.push_back(to);
        else dq.push_front(to);
      }
    }
  }
```

```
  return min(dis[n], dis[n+n]);
}

int main()
{
  cin >> n >> m >> K;
  while (m -- ) {
    int a, b, c; cin >> a >> b >> c;
    if (c == 1) vec[a].push_back({b,1}), vec[b].push_back({a,1});
    else vec[n+a].push_back({n+b,1}), vec[n+b].push_back({n+a,1});
  }
  while (K -- ) {
    int x; cin >> x;
    vec[x].push_back({n+x,0}), vec[n+x].push_back({x,0});
  }

  int res = bfs();
  cout << (res==inf ? -1 : res) << endl;
  return 0;
}
```

blackout

先把所有数据输入, 然后反过来考虑, 用并查集维护

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 5e5 + 5;
int u[maxn], v[maxn], w[maxn];
bool st[maxn];

int f[maxn], sz[maxn];
bool is_ok[maxn];
int fFind(int x) {
  if (f[x] != x) f[x] = fFind(f[x]);
  return f[x];
}

int solve(int a, int b) {
  int fa = fFind(a), fb = fFind(b), sum = 0;
  if (fa != fb) {
    if (is_ok[fa] && !is_ok[fb]) sum += sz[fb];
    if (!is_ok[fa] && is_ok[fb]) sum += sz[fa];

    f[fa] = fb; sz[fb] += sz[fa];
    if (is_ok[fa]) is_ok[fb] = true;
  }
  return sum;
}
```

```
int ans[maxn];

int main()
{
  int n, m, E; cin >> n >> m >> E;
  for (int i = 1; i <= E; ++i) cin >> u[i] >> v[i];
  int Q; cin >> Q;
  for (int i = 1; i <= Q; ++i) cin >> w[i], st[w[i]] = true;

  for (int i = 1; i <= n+m; ++i) f[i] = i, sz[i] = (i<=n), is_ok[i] = (i>n);

  int res = 0;
  for (int i = 1; i <= E; ++i) {
    if (st[i]) continue;
    int a = u[i], b = v[i];
    res += solve(a, b);
  }

  for (int i = Q; i >= 1; --i) {
    ans[i] = res;
    int a = u[w[i]], b = v[w[i]];
    res += solve(a, b);
  }

  for (int i = 1; i <= Q; ++i) cout << ans[i] << endl;
  return 0;
}
```

Well-defined Path Queries on a Namori

先拓扑排序找环, 然后按照环上的点进行分组

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
vector<int> vec[maxn];
int deg[maxn], f[maxn];
int n;

void topo_sort() {
  queue<int> q;
  for (int i = 1; i <= n; ++i) {
    if (deg[i] == 1) q.push(i);
  }

  while (!q.empty()) {
    int u = q.front(); q.pop();
    for (int i : vec[u]) {
```

```
        --deg[i];
        if (deg[i] == 1) q.push(i);
      }
    }
  }

  void dfs(int u, int val) {
    if (f[u]) return;
    f[u] = val;
    for (int i : vec[u]) {
      if (deg[i] != 2) dfs(i, val);
    }
  }

  int main()
  {
    cin >> n;
    for (int i = 1; i <= n; ++i) {
      int a, b; cin >> a >> b; ++deg[a], ++deg[b];
      vec[a].push_back(b), vec[b].push_back(a);
    }

    topo_sort();

    for (int i = 1; i <= n; ++i) {
      if (deg[i] == 2) dfs(i, i);
    }

    int Q; cin >> Q;
    while (Q -- ) {
      int a, b; cin >> a >> b;
      if (f[a] == f[b]) cout << "Yes" << endl;
      else cout << "No" << endl;
    }
    return 0;
  }
```